

Sistem ve Network Atölyesi

Secure Access & File Sharing (Güvenli Erişim ve Dosya Paylaşımı)

Ajanda

- Temel Linux Bilgileri
 - How to linux Works
 - Package Managers
 - Temel komutlar
 - Linux SSH Key Yönetimi
- SSL/TLS Nedir?
- SCP, Rsync, SFTP ile Güvenli Dosya Transferi
- NFS (Network File System)
- SSH Tunneling ile Port Yönlendirme
- Hands-on
 - SSH Key yönetimi
 - Nginx + SSL Kurulumu
 - SCP & Rsync ile Güvenli Dosya Transferi
 - NFS Server Kurulumu
 - SSH Tunnel ile MySQL Portuna Güvenli Erişim

Temel Linux Bilgileri

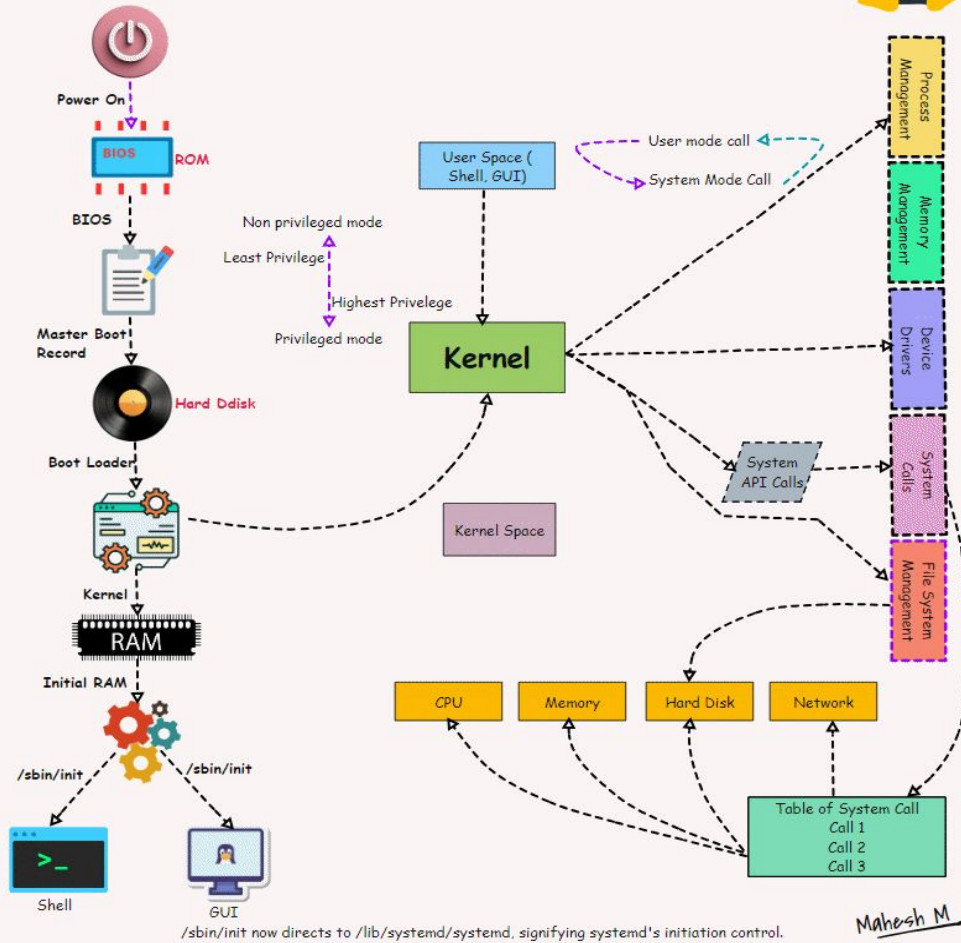


Neden Linux Öğrenmeliyim?

- **AWS EC2 instance'larının %76'sı** → Linux çalıştırıyor
- **Fortune 500 şirketlerinin %90'ı** → Linux kullanıyor
- **Modern DevOps/Cloud akışlarının omurgası** → Linux.
- **Docker, Kubernetes, CI/CD** → Hepsi Linux tabanlı
- **Günlük görevler:** bağlan, log bak, servis yönet, dağıtım yap.



HOW LINUX WORKS



Temel Kavramlar:

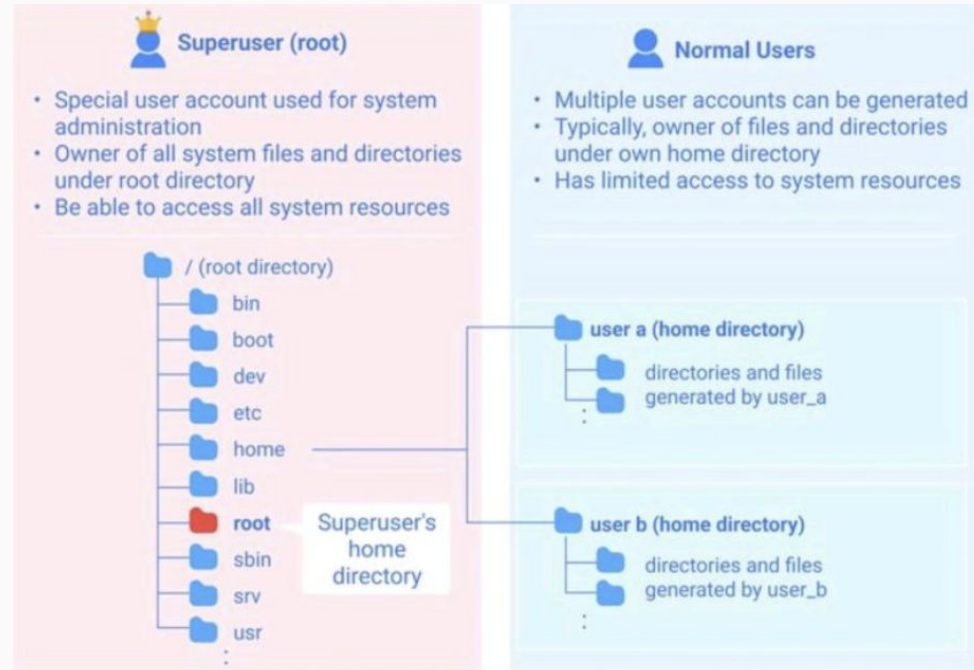
- **Kernel:** Donanım ile yazılım arası köprü, kaynak yöneticisi
- **Shell:** Kullanıcı ile kernel arası arayüz (bash, zsh)
- **Process:** Çalışan program instance'ı
- **User/Group:** Erişim kontrolü için kimlik sistemi

Root vs Normal User

Root vs Normal User

- **root**: sistemin her şeyine erişir (tehlike + sorumluluk).
- **normal user**: kendi alanı (~) ve yetkileriyle sınırlı.

Command	Description
#	root user
\$	Normal user
/	root directory
~	home directory



sudo Nedir? Ne Zaman Kullanılır

Superuser do: Komutu **geçici** olarak root yetkisiyle çalıştırır.

- **Kullan:**

- Sistem dosyası / servis: `/etc/*`, `systemctl`, `apt/yum/dnf`
- Düşük port (<1024) isteyen servisler (80/443 vb.)
- Paket kur/kaldır, kernel modülü, daemon ayarı

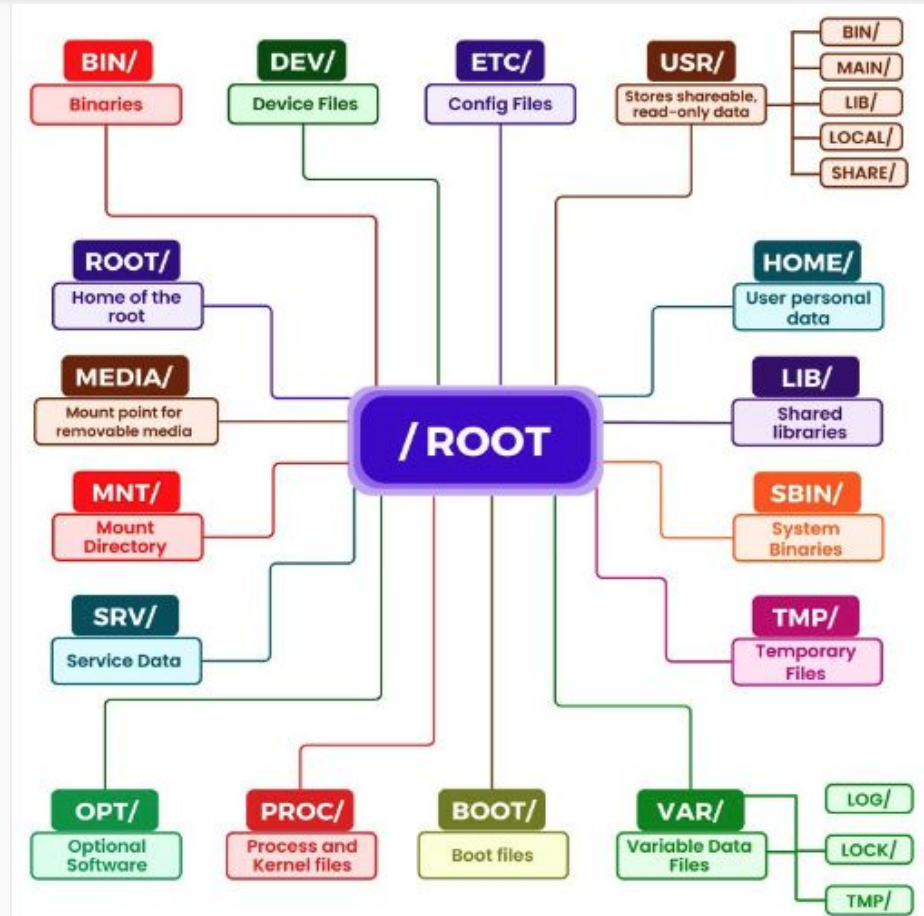
- **Kullanma:**

- Home dizini işlemleri (`~/proj`, `~/logs`)
- Okuma işlemleri (log görüntüleme genelde root istemez)
- “Sorun var → her komuta sudo” **yanlış refleks**

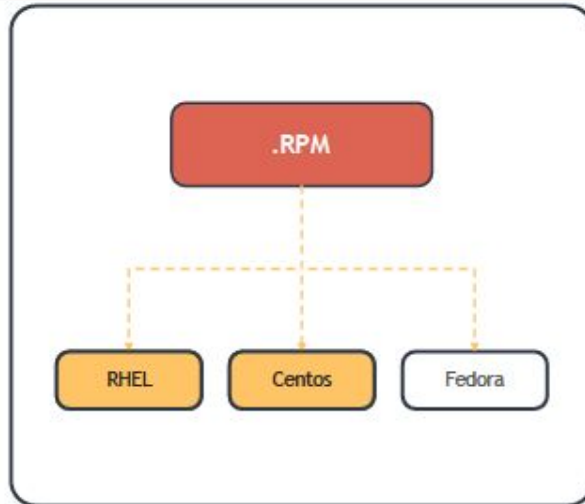
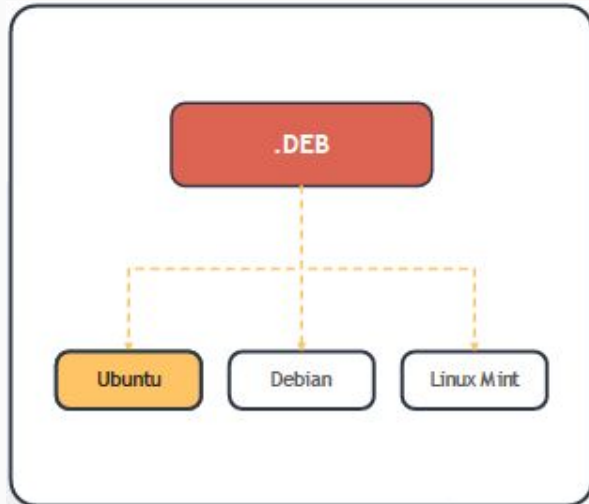
Tehlikeli sudo örnekleri

- `sudo rm -rf /` 
- `sudo chmod -R 777 /` 
- `sudo chown -R root:root ~` 

File system hierarchy



Package Managers



Package Managers

YUM Package Manager

Software Repository



YUM Package Manager



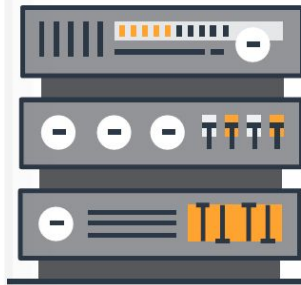
Local or Remote
(HTTP/HTTPS/FTP)

```
/etc/yum.repos.d
```

```
/etc/yum.repos.d/redhat.repo
```

```
/etc/yum.repos.d/nginx.repo
```

RPM Package Manager



```
[~]$ yum repolist
```

```
[~]$ yum provides scp
```

```
[~]$ yum update
```

```
[~]$ yum remove httpd
```

```
[~]$ yum update telnet
```

```
[~]$ yum install httpd
```

Working with RPM

Installation

```
[~]$ rpm -ivh telnet.rpm
```

Uninstalling

```
[~]$ rpm -e telnet.rpm
```

Upgrade

```
[~]$ rpm -Uvh telnet.rpm
```

Query

```
[~]$ rpm -q telnet.rpm
```

Verifying

```
[~]$ rpm -Vf <path to file>
```

Package Managers

Working with DPKG

Installation / Upgrade

```
[~]$ dpkg -i telnet.deb
```

Uninstalling

```
[~]$ dpkg -r telnet.deb
```

List

```
[~]$ dpkg -l telnet
```

Status

```
[~]$ dpkg -s telnet
```

Verifying

```
[~]$ dpkg -p <path to file>
```

```
[~]$ apt update
```

```
[~]$ apt upgrade
```

```
[~]$ apt edit-sources
```

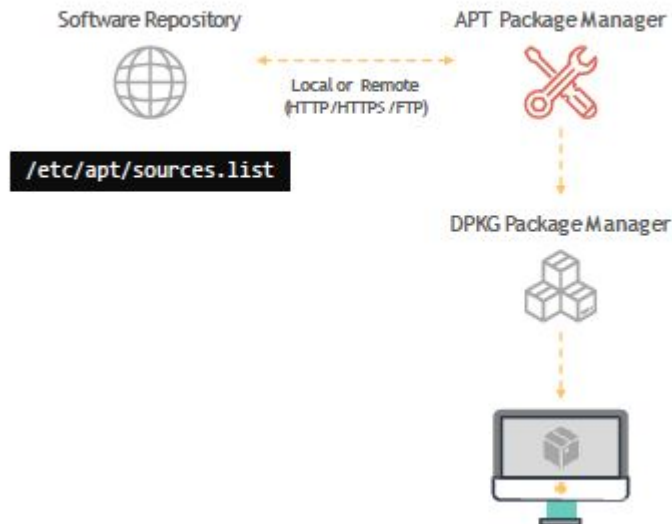
```
[~]$ apt install telnet
```

```
[~]$ apt remove telnet
```

```
[~]$ apt search telnet
```

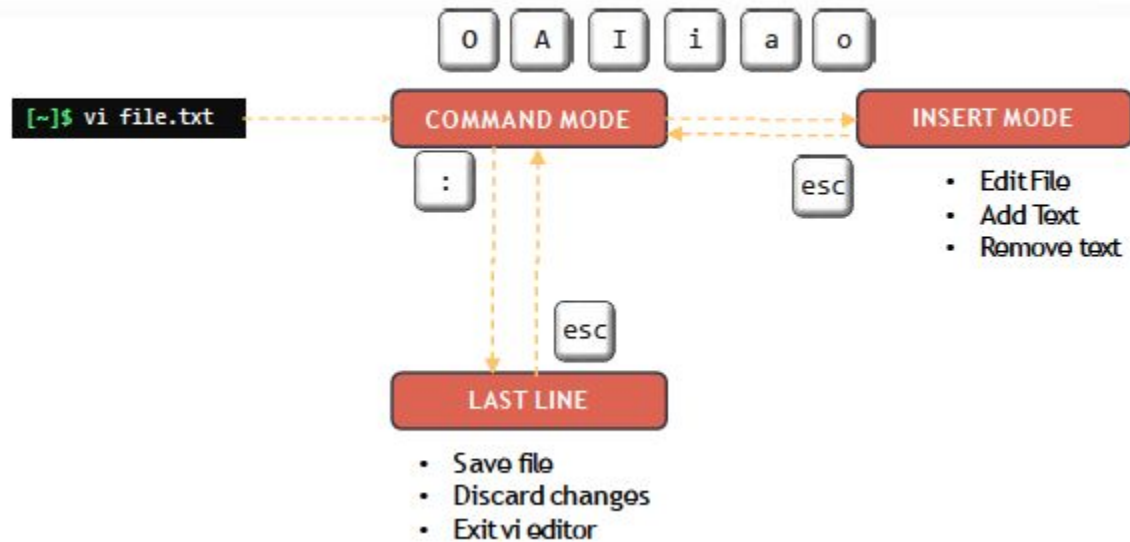
```
[~]$ apt list | grep telnet
```

APT



Vi Editor

VI EDITOR MODES



Vi command mode

Move Around

Copy a Line

Paste

Delete a letter

^

< v >

K

H J L

y y

p

x

The diagram illustrates a text editor interface with a vertical list of command buttons on the left and a sequence of tokens on the right. The buttons are: "Delete a line", "Delete 3 lines", "Undo", and "Redo". The tokens are: "d", "d", "d", "3", "d", "u", and "r". The tokens are arranged in a sequence that corresponds to the commands: "Delete a line" (d, d), "Delete 3 lines" (d, 3, d), "Undo" (u), and "Redo" (r).

Command	Token(s)
Delete a line	d, d
Delete 3 lines	d, 3, d
Undo	u
Redo	r

```
This is the first line.  
Followed by the second line.  
Third line is very long compared to the previous two lines.  
Hello there!  
hello there!  
  
-- INSERT --
```



The diagram illustrates Emacs navigation keys. It features a large, light gray, rounded rectangular background. On this background, there are six red rectangular buttons with rounded corners, arranged in two rows of three. The top row contains buttons labeled 'Find', '/line', and '?line'. The bottom row contains buttons labeled 'Find Next', 'n', and 'N'. The buttons are arranged in a grid-like fashion, with the red buttons on the left and the white buttons on the right.

Find	/line	?line
Find Next	n	N

INSERT and LASTLINE MODE

INSERT MODE

Insert Mode

i,o,a l,O,A

Command Mode

ESC

```
This is the first line.  
Followed by the second line.  
Third line is very long compared to the previous two lines.  
Hello there!  
hello there!
```

```
-- INSERT --
```

LASTLINE MODE

Save

W

Quit

q:

Save & Quit

WQ

Quit

Without Confirmation

g!

```
This is the first line.  
Followed by the second line.  
Third line is very long compared to the previous two lines.  
Hello there!  
Hello there!  
I made some changes to this file.
```

--
--
--
--
--
--
--
--
--
--
--
--

```
:w                               1,1          A!
```


Basic Linux commands

1 - SYSTEM INFORMATION

<code>uname -a</code>	# Display Linux system information
<code>uname -r</code>	# Display kernel release information
<code>cat /etc/redhat-release</code>	# Show which version of redhat installed
<code>uptime</code>	# Show how long the system has been running + load
<code>hostname</code>	# Show system host name
<code>hostname -I</code>	# Display the IP addresses of the host
<code>last reboot</code>	# Show system reboot history
<code>date</code>	# Show the current date and time
<code>cal</code>	# Show this month's calendar
<code>w</code>	# Display who is online
<code>whoami</code>	# Who you are logged in as

Basic Linux commands

2 - HARDWARE INFORMATION

<code>dmesg</code>	# Display messages in kernel ring buffer
<code>cat /proc/cpuinfo</code>	# Display CPU information
<code>cat /proc/meminfo</code>	# Display memory information
<code>free -h</code>	# Display free and used memory (-h for human readable -m for MB, -g for GB.)
<code>lspci -tv</code>	# Display PCI devices
<code>lsusb -tv</code>	# Display USB devices
<code>dmidecode</code>	# Display DMI/SMBIOS (hardware info) from the BIOS
<code>hdparm -i /dev/sda</code>	# Show info about disk sda
<code>hdparm -tT /dev/sda</code>	# Perform a read speed test on disk sda
<code>badblocks -s /dev/sda</code>	# Test for unreadable blocks on disk sda

Basic Linux commands

3 - PERFORMANCE MONITORING AND STATISTICS

<code>top</code>	# Display and manage the top processes
<code>htop</code>	# Interactive process viewer (top alternative)
<code>mpstat 1</code>	# Display processor related statistics
<code>vmstat 1</code>	# Display virtual memory statistics
<code>iostat 1</code>	# Display I/O statistics
<code>tail 100 /var/log/messages</code>	# Display the last 100 syslog messages (Use /var/log/syslog for Debian based systems.)
<code>tcpdump -i eth0</code>	# Capture and display all packets on interface eth0
<code>tcpdump -i eth0 'port 80'</code>	# Monitor all traffic on port 80 (HTTP)
<code>lsof</code>	# List all open files on the system
<code>lsof -u user</code>	# List files opened by user
<code>free -h</code>	# Display free and used memory (-h for human readable, -m for MB, -g for GB.)

Basic Linux commands

4 - USER INFORMATION AND MANAGEMENT

<code>id</code>	<code># Display the user and group ids of your current user.</code>
<code>last</code>	<code># Display the last users who have logged onto the system.</code>
<code>who</code>	<code># Show who is logged into the system.</code>
<code>w</code>	<code># Show who is logged in and what they are doing.</code>
<code>groupadd test</code>	<code># Create a group named "test".</code>
<code>useradd -c "John Smith" -m john</code>	<code># Create an account named john, with a comment of "John Smith" and create the user's home directory.</code>
<code>userdel john</code>	<code># Delete the john account.</code>
<code>usermod -aG sales john</code>	<code># Add the john account to the sales group</code>

Basic Linux commands

5 - FILE AND DIRECTORY COMMANDS

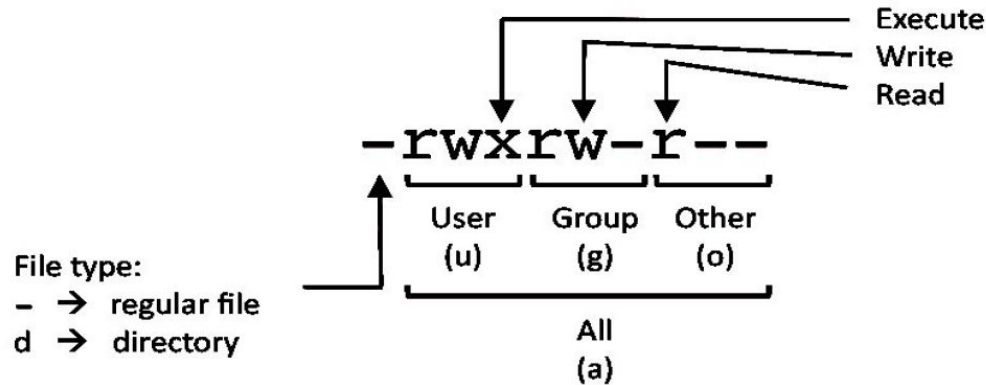
<code>ls -al</code>	# List all files in a long listing (detailed) format
<code>pwd</code>	# Display the present working directory
<code>mkdir directory</code>	# Create a directory
<code>rm file</code>	# Remove (delete) file
<code>rm -r directory</code>	# Remove the directory and its contents recursively
<code>rm -f file</code>	# Force removal of file without prompting for confirmation
<code>rm -rf directory</code>	# Forcefully remove directory recursively
<code>cp file1 file2</code>	# Copy file1 to file2
<code>cp -r source_directory destination</code>	# Copy source_directory recursively to destination. If destination exists, copy source_directory into destination, otherwise create destination with the contents of source_directory.
<code>mv file1 file2</code>	# Rename or move file1 to file2. If file2 is an existing directory, move file1 into directory file2
<code>ln -s /path/to/file linkname</code>	# Create symbolic link to linkname
<code>touch file</code>	# Create an empty file or update the access and modification times of file.
<code>cat file</code>	# View the contents of file
<code>less file</code>	# Browse through a text file
<code>head file</code>	# Display the first 10 lines of file
<code>tail file</code>	# Display the last 10 lines of file
<code>tail -f file</code>	# Display the last 10 lines of file and "follow" the file as it grows.

6 - PROCESS MANAGEMENT

<code>ps</code>	# Display your currently running processes
<code>ps -ef</code>	# Display all the currently running processes on the system.
<code>ps -ef grep processname</code>	# Display process information for processname
<code>top</code>	# Display and manage the top processes
<code>htop</code>	# Interactive process viewer (top alternative)
<code>kill pid</code>	# Kill process with process ID of pid
<code>killall processname</code>	# Kill all processes named processname
<code>program &</code>	# Start program in the background
<code>bg</code>	# Display stopped or background jobs
<code>fg</code>	# Brings the most recent background job to foreground
<code>fg n</code>	# Brings job n to the foreground

Basic Linux commands

7 - FILE PERMISSIONS



PERMISSION

EXAMPLE

U	G	W	
rwx	rwx	rwx	chmod 777 filename # Use sparingly!
rwx	rwx	r-x	chmod 775 filename
rwx	r-x	r-x	chmod 755 filename
rw-	rw-	r--	chmod 664 filename
rw-	r--	r--	chmod 644 filename

LEGEND

U = User

G = Group

W = World

r = Read

w = write

x = execute

- = no access

Basic Linux commands

8 - NETWORKING

<code>ifconfig -a</code>	# Display all network interfaces and ip address
<code>ifconfig eth0</code>	# Display eth0 address and details
<code>ethtool eth0</code>	# Query or control network driver and hardware settings
<code>ping host</code>	# Send ICMP echo request to host
<code>whois domain</code>	# Display whois information for domain
<code>dig domain</code>	# Display DNS information for domain
<code>dig -x IP_ADDRESS</code>	# Reverse lookup of IP_ADDRESS
<code>host domain</code>	# Display DNS ip address for domain
<code>hostname -i</code>	# Display the network address of the host name.
<code>hostname -I</code>	# Display all local ip addresses
<code>wget http://domain.com/file</code>	# Download http://domain.com/file
<code>netstat -nutlp</code>	# Display listening tcp and udp ports and corresponding programs

Basic Linux commands

9 - ARCHIVES (TAR FILES)

<code>tar cf archive.tar directory</code>	<code># Create tar named archive.tar containing directory.</code>
<code>tar xf archive.tar</code>	<code># Extract the contents from archive.tar.</code>
<code>tar czf archive.tar.gz directory</code>	<code># Create a gzip compressed tar file name</code> <code>...</code>
<code>tar xzf archive.tar.gz</code>	<code># Extract a gzip compressed tar file.</code>
<code>tar cjf archive.tar.bz2 directory</code>	<code># Create a tar file with bzip2 compression</code>
<code>tar xjf archive.tar.bz2</code>	<code># Extract a bzip2 compressed tar file.</code>

Basic Linux commands

10 - INSTALLING PACKAGES

<code>yum search keyword</code>	<code># Search for a package by keyword.</code>
<code>yum install package</code>	<code># Install package.</code>
<code>yum info package</code>	<code># Display description and summary information about package.</code>
<code>rpm -i package.rpm</code>	<code># Install package from local file named package.rpm</code>
<code>yum remove package</code>	<code># Remove/uninstall package.</code>
<code>tar zxvf sourcecode.tar.gz</code> <code>cd sourcecode</code> <code>./configure</code> <code>make</code> <code>make install</code>	<code># Install software from source code.</code>

Basic Linux commands

11 - SEARCH

<code>grep pattern file</code>	<code># Search for pattern in file</code>
<code>grep -r pattern directory</code>	<code># Search recursively for pattern in directory</code>
<code>locate name</code>	<code># Find files and directories by name</code>
<code>find /home/john -name 'prefix*'</code>	<code># Find files in /home/john that start with "prefix".</code>
<code>find /home -size +100M</code>	<code># Find files larger than 100MB in /home</code>

12 - SSH LOGINS

<code>ssh host</code>	<code># Connect to host as your local username.</code>
<code>ssh user@host</code>	<code># Connect to host as user</code>

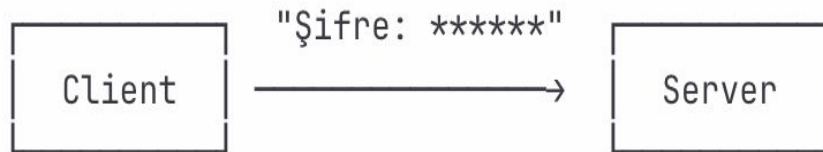
linux komutları pratik



Linux SSH Key Yönetimi

Şifre vs SSH Key Authentication

Şifre ile Giriş (Eski Yöntem):



Sorunlar:

- ❌ Her seferinde şifre yaz
- ❌ Brute-force saldırılarına açık
- ❌ Otomasyon yapılamaz (script'lerde şifre saklanamaz)
- ❌ 50 server varsa 50 şifre mi ezberleyeceksin?

SSH Key ile Giriş (Modern Yöntem):



Avantajlar:

- ✅ Şifre yazmaya gerek yok (passwordless)
- ✅ Çok daha güvenli (4096-bit şifreleme)
- ✅ Brute-force saldırısı neredeyse imkansız
- ✅ Otomasyon kolay (CI/CD, cron job'lar)
- ✅ Tek key ile 100 server'a giriş

Public vs Private Key Mantığı

🔓 **Public Key = Kilit (Asma kilit)**

- **Server'lara kopyalyorsun**
- **Herkes görebilir, sorun değil**
- **Dosya: ~/.ssh/id_rsa.pub**

🔑 **Private Key = Anahtar**

- **Sadece sende**
- **KİMSEYLE PAYLAŞMA!**
- **Kaybolursa yenisini yap**
- **Dosya: ~/.ssh/id_rsa**



Çalışma Prensipleri:

1. Private key ile bir mesaj imzalyorsun
2. Server public key ile imzayı doğruluyor
3. Eşleşiyorsa → Sen gerçekten sensin → Giriş izni!

Linux SSH Key Yönetimi

Key Pair oluşturma:

Modern ve en güvenli: `ssh-keygen -t ed25519 -C "email@example.com"`

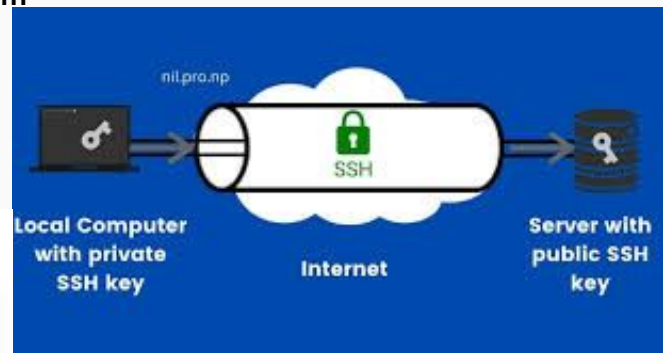
Eski sistemlerle uyumluluk için: `ssh-keygen -t rsa -b 4096 -C "email@example.com"`

Public key → sunucuya `~/.ssh/authorized_keys` içine eklenir

Private key → yalnızca kullanıcıda kalır, kimseyle paylaşılmaz

SSH Key Türleri

Algoritma	Bit Uzunluğu	Güvenlik	Hız	Önerilen
RSA	2048/4096	Yüksek	Orta	✓ Evet
Ed25519	256	Çok Yüksek	Hızlı	✓✓ En iyi
ECDSA	256	Yüksek	Hızlı	⚠ RSA/Ed25519 tercih et
DSA	1024	Düşük	Hızlı	✗ Eski, kullanma



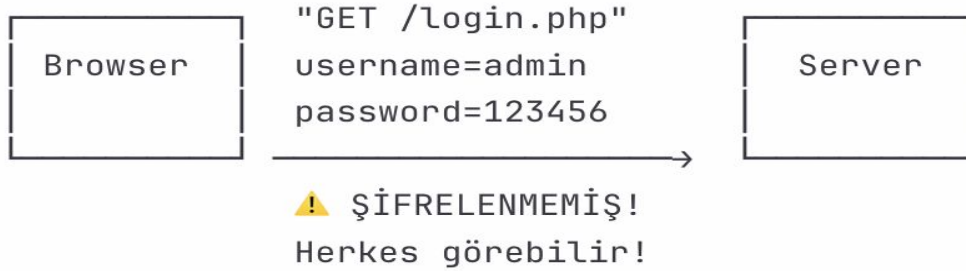
Linux SSH Key Yönetimi



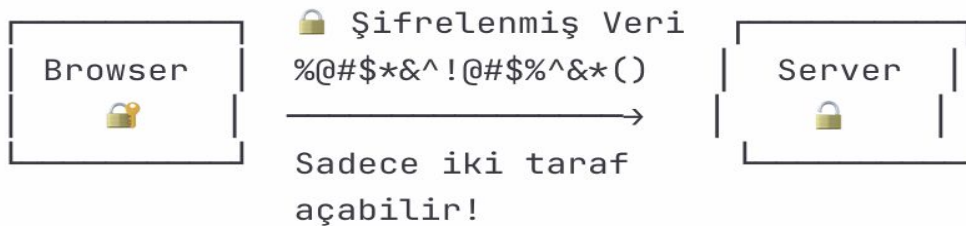
SSL/TLS

HTTP vs HTTPS

HTTP (Hypertext Transfer Protocol):



HTTPS (HTTP + SSL/TLS):



SSL Sertifikası Nedir?

Pasaport Benzetmesi:

- **Pasaport:** Kimliğini kanıtlar → Güvenilir otorite (Devlet) onaylar
- **SSL Sertifikası:** Web sitesinin kimliğini kanıtlar → Certificate Authority (CA) onaylar

Sertifika İçindekiler:

- Alan adı (Domain): example.com
- Şirket bilgileri: Example Inc.
- Geçerlilik tarihi: 2025-01-01 → 2026-01-01
- Public key: (şifreleme için)
- CA imzası: (güvenilirlik için)

SSL/TLS: Veriyi şifreleyerek güvenli iletişim sağlar

HTTPS, SMTPS, IMAPS gibi protokollerde kullanılır

Sertifika yönetimi:

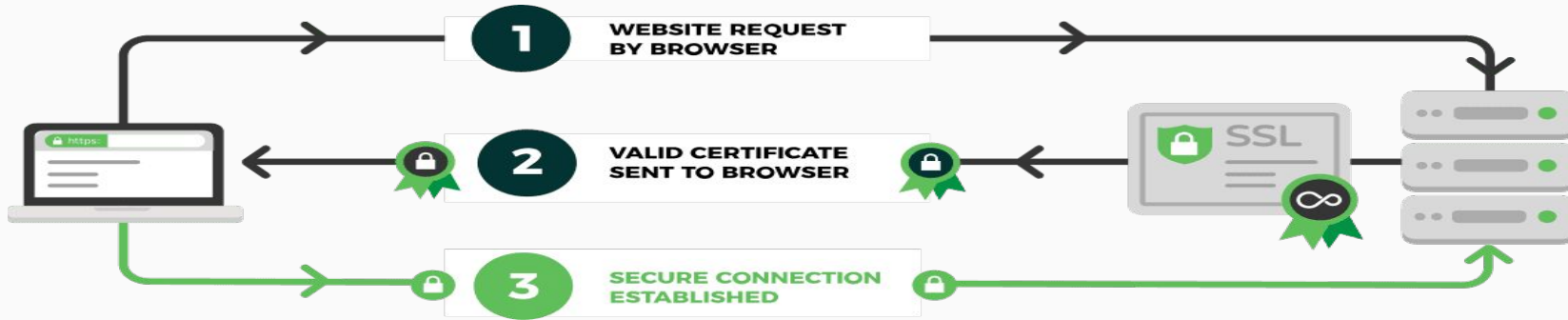
- **AWS ACM (Certificate Manager):** SSL sertifikalarını yönetir
- **Let's Encrypt:** Ücretsiz SSL sertifikaları sağlar

Sertifika Türleri

Tür	Maliyet	Süre	Güvenilirlik	Kullanım
Self-Signed	\$0	Sınırsız	⚠ Tarayıcı uyarı	Test/Geliştirme
Let's Encrypt	\$0	90 gün (otomatik yenileme)	✅ Güvenilir	Küçük/orta siteler
Commercial CA	\$50-500/yıl	1-3 yıl	✅✅ Çok güvenilir	Kurumsal

SSL/TLS Handshake (Nasıl Çalışır?)

1. **Client:** "Merhaba, HTTPS ile bağlanmak istiyorum"
- ↓
2. **Server:** "Tamam, işte sertifikam (public key dahil)"
- ↓
3. **Client:** Sertifikayı CA ile doğrula (Geçerli mi? Süresi dolmamış mı? Domain eşleşiyor mu?)
- ↓
4. **Client:** Rastgele şifreleme anahtarı üret Server'ın public key ile şifrele Server'a gönder
- ↓
5. **Server:** Private key ile aç, anahtarı öğren
- ↓
6. 🔒 Artık iki taraf da aynı anahtarı biliyor
- ↓
7. Tüm iletişim bu anahtar ile şifrelenir



NGINX SSL KURULUMU



Dosya Transferleri

SCP, Rsync, SFTP ile Güvenli Dosya Transferi

SCP (Secure Copy):

- `scp file.txt user@server:/path/`

Rsync:

- Artımlı dosya senkronizasyonu
- `rsync -avz file user@server:/path/`

SFTP (SSH File Transfer Protocol):

- FTP'ye benzer ama SSH ile şifrelenmiş bağlantı



SCP vs Rsync vs SFTP - Hangisini Ne Zaman?

- # Tek dosya hızlıca transfer : scp file.txt user@server:/path/
- # Büyük dizin ilk kez kopyala : rsync -avz dir/ user@server:/path/
- # Günlük backup (sadece değişenler): rsync -avz --delete src/ dest/
- # İnteraktif dosya yönetimi (FTP gibi): sftp user@server
- # Production deployment : rsync -avz --exclude node_modules/ app/ server:/var/www/

- ✓ SCP
- ✓ Rsync
- ✓ Rsync
- ✓ SFTP
- ✓ Rsync

Özellik	SCP	Rsync	SFTP
Hız	Hızlı	Çok Hızlı (delta transfer)	Orta
Artımlı Sync	✗	✓	✗
İnteraktif	✗	✗	✓
Sıkıştırma	⚠	✓	✗
Resume	✗	✓	⚠
Kullanım	Tek dosya	Backup/Sync	FTP yerine

Dosya Transferleri (scp rsync sftp)



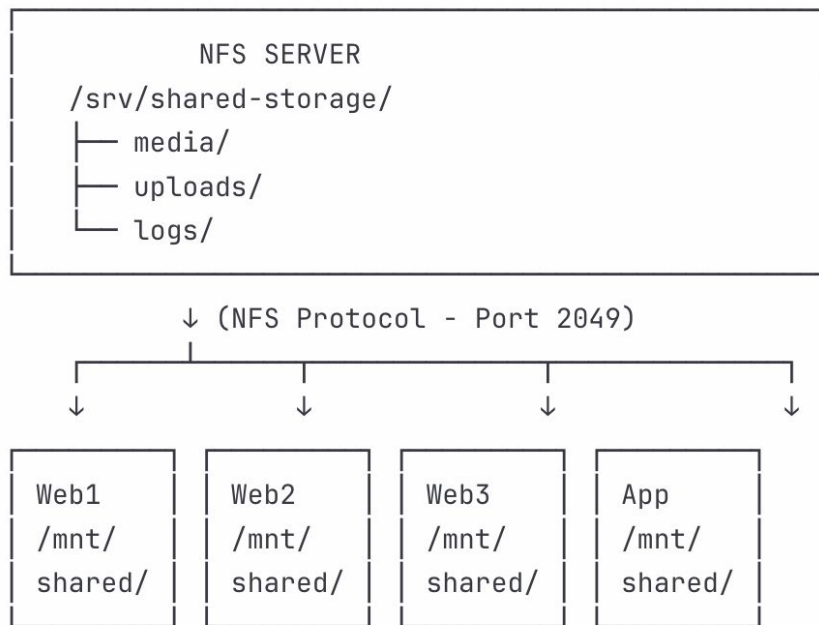
NFS

NFS (Network File System) Nedir?

Basit Tanım:

Bir sunucudaki dizini, diğer sunuculara "sanki local disk" gibi bağlama

NFS Nasıl Çalışır?



↑ Hepsi aynı dosyaları görüyor!

NFS Kullanım Senaryoları:

1. Web Sunucuları (Ortak Media)
2. Kubernetes Persistent Volume
3. Log Toplama

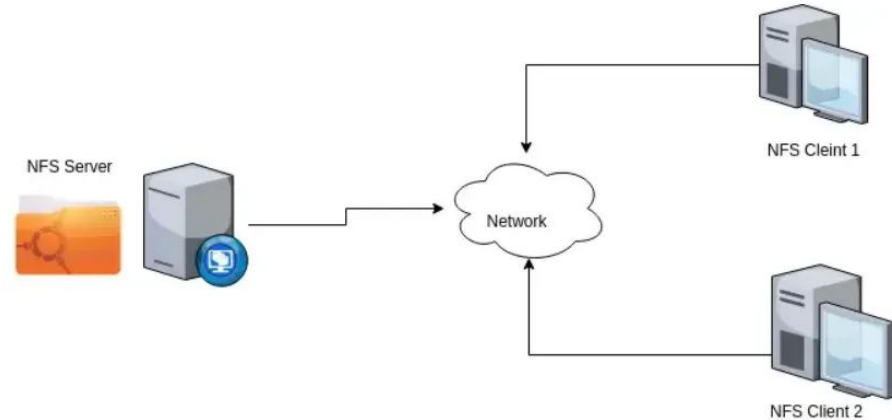
Dezavantaj:

Şifreleme yok → güvenlik için VPN/SSH tüneliyle kullanılmalı

NFS Güvenlik - ÖNEMLİ!

NFS şifreleme yapmaz!

- ❌ Internet üzerinden kullanma
- ❌ Public network'te açma
- ✅ Private VPC içinde kullan
- ✅ Security Group ile sınırla
- ✅ SSH tunnel ile erişersen daha güvenli



NFS vs Alternatifler

Özellik	NFS	AWS EFS	S3	Samba (CIFS)
Platform	Linux	AWS	Tümü	Windows+Linux
Performans	Yüksek	Orta	Düşük	Orta
Maliyet	Ücretsiz	\$\$	\$	Ücretsiz
Şifreleme	❌	✅	✅	⚠️
Kurulum	Kolay	Yönetilmiyor	API	Orta

NFS Server Kurulumu



SSH TUNNEL

SSH Tunnel Nedir?

Basit Tanım:SSH bağlantısı içinde başka bir bağlantı açma (tünel)

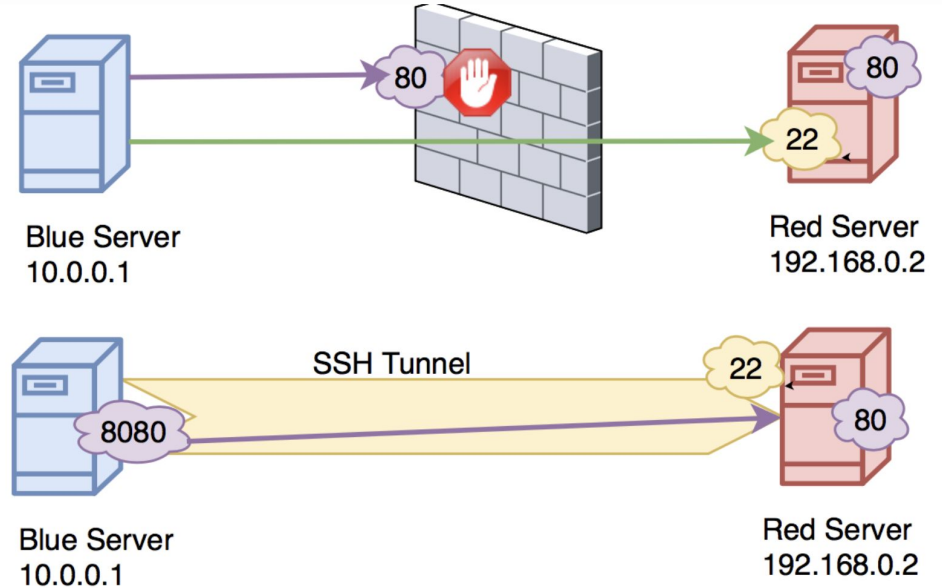
Analoji:

Tünel (tunnel) = Boru içinden boru

SSH bağlantısı = Dış boru (şifreli)

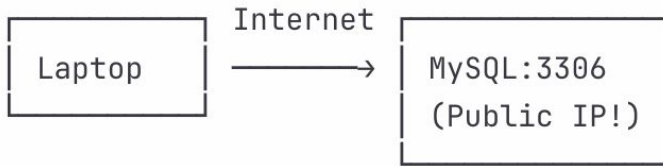
İçindeki bağlantı = İç boru (MySQL, HTTP, vb.)

Local Port Forward



Neden SSH Tunnel?

❌ YANLIŞ YÖNTEM:



⚠️ Güvenlik riski!

⚠️ Brute-force saldırılarına açık

⚠️ Veri şifrelenmemiş

✅ DOĞRU YÖNTEM (SSH Tunnel):



✅ Şifreli bağlantı

✅ Public'e kapalı

✅ Bastion üzerinden kontrollü erişim

SSH Tunneling ile Port Yönlendirme

Local Port Forwarding:

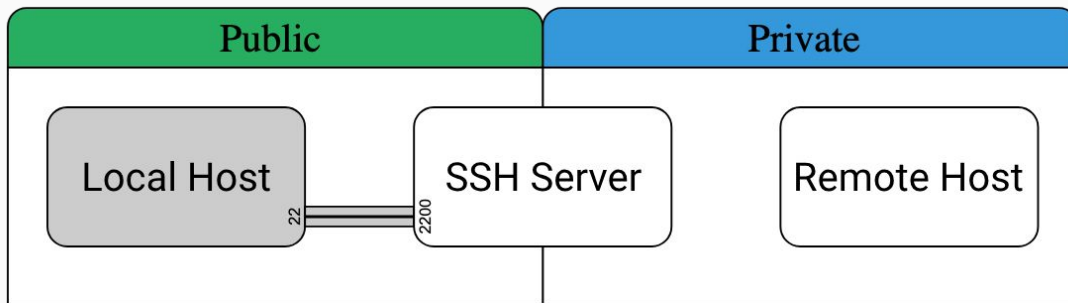
- `ssh -L 3306:localhost:3306 user@bastion`
- → Yerel makineden MySQL'e güvenli erişim

Remote Port Forwarding:

- `ssh -R 8080:localhost:80 user@server`
- → Uzak tarafta kendi portunu açıp yönlendirme

Kullanım alanları:

- MySQL, PostgreSQL erişimi
- HTTP servislerine güvenli erişim



SSH TUNNEL TÜRLERİ



1. Local Port Forwarding (-L)

Kullanım: Uzak servise local'den erişmek istiyorsun

```
ssh -L [local_port]:[remote_host]:[remote_port] user@jump_host
```

Örnek:

```
ssh -L 3306:db-server:3306 ec2-user@bastion
```

Ne olur?

localhost:3306 → bastion → db-server:3306

Senaryo:

Laptop'tan private MySQL'e bağlanmak istiyorsun:

1. SSH tunnel aç: `ssh -L 3306:10.0.2.50:3306 bastion`
2. MySQL client: `mysql -h localhost -P 3306`
3. Bağlantı: laptop → bastion → MySQL

2. Remote Port Forwarding (-R)

Kullanım: Local servisi uzaktan erişilebilir yapmak istiyorsun

```
ssh -R [remote_port]:localhost:[local_port] user@remote_host
```

Örnek:

```
ssh -R 8080:localhost:3000 ec2-user@public-server
```

Ne olur?

public-server:8080 → SSH tunnel → laptop:3000

Senaryo:

Laptop'ta geliştirme yapıyorsun (port 3000) Arkadaşına göstermek istiyorsun:

1. SSH tunnel aç: `ssh -R 8080:localhost:3000 server`
2. Arkadaşın browser'da: `http://server-ip:8080`
3. Laptop'taki uygulamayı görür!

3. Dynamic Port Forwarding (-D) - SOCKS Proxy

Kullanım: Tüm trafiği SSH üzerinden yönlendir

```
ssh -D 1080 user@jump_host
```

Browser'da SOCKS proxy ayarla: localhost:1080

Artık tüm web trafiğin jump_host üzerinden gider

Senaryo:

Güvensiz Wi-Fi'desin (havaalanı, kafe) Tüm trafiğini şifrele:

1. SSH tunnel: `ssh -D 1080 your-server`
2. Browser → SOCKS proxy: localhost:1080
3. Tüm trafik şifreli!

SSH Tunnel ile MySql portuna güvenli erişim





Ek Kaynaklar ve Öğrenmeye Devam

Pratik Yapabileceğin Siteler

1. OverTheWire - Bandit <https://overthewire.org/wargames/bandit/> → Linux komutları öğren (oyun formatında)

Host: bandit.labs.overthewire.org

Port: 2220

Username: bandit0

Password: bandit0

`ssh bandit0@bandit.labs.overthewire.org -p 2220`

2. Linux Journey <https://linuxjourney.com/> → Adım adım Linux eğitimi

3. Redhat <https://www.redhat.com/en/interactive-labs> → Red Hat interactive labs

4. KodeKloud <https://kodekloud.com/free-labs/linux> → Linux Free Labs

5. AWS Free Tier <https://aws.amazon.com/free/> → Kendi lab ortamını kur